MODULE 1 · WINDSHEAR

Lesson 02: Avoidance and Recognition

Lesson 2 of 5 · about 7 min

LEARNING OBJECTIVES

- Use weather reports and alerts to avoid windshear.
- Recognise the cues of a windshear encounter.
- Understand predictive and reactive windshear systems.



Avoid: delay the approach or divert.

Avoid — the best defence

- Use the METAR, TAF, SIGMET, ATIS and pilot reports: `TS`, `+TSRA`, `CB`, gusts, `WS` groups in METARs (e.g. `WS R26`) and low-level windshear alerts.
- Watch for visual signs: virga, rain shafts, dust rings on the ground, roll clouds and rapidly changing wind socks.
- Delay the take-off or approach when severe windshear or a microburst is reported or forecast, or when a strong thunderstorm is over or near the aerodrome. Hold or divert.
- If windshear is possible: use the maximum take-off thrust or the take-off settings recommended by your FCOM/AOM, consider the longest suitable runway, and brief the escape manoeuvre.

Recognise

- Unexpected **airspeed changes** — typically more than about 15 kt.
- **Vertical speed** deviations — for example 500 ft/min or more from normal.
- Pitch attitude changes and deviations from the glide path.
- Unusual thrust lever positions needed to hold the speed.

- Aural and visual windshear warnings.

Windshear systems

- **Predictive windshear** (weather radar based, where fitted) looks ahead and alerts before the aircraft enters the windshear — “WINDSHEAR AHEAD” cautions and warnings, depending on the system.
- **Reactive windshear** (EGPWS or flight control computer based, where fitted) alerts when the aircraft is already in windshear — “WINDSHEAR, WINDSHEAR, WINDSHEAR”.
- Know which systems your aircraft has and how they are inhibited during certain phases.

FLEET NOTE

The available windshear warnings differ between the B737-800, the Q400 and the Q200. Check the FCOM / AOM for your aircraft.

KEY TAKEAWAYS

- Avoid first: reports, radar, visual signs; delay or divert.
- Cues: speed changes over about 15 kt, vertical speed deviations of about 500 ft/min, unusual thrust.
- Predictive warns before entry; reactive warns when inside.

KNOWLEDGE CHECK

What is the most effective defence against microburst windshear?

A. High approach speed

B. Avoidance — delay the take-off or approach

C. Flaps up landing

D. Autopilot off

Answer: B. Avoidance is the most effective defence.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.